

A Experimental design

Each condition: 30 independent runs $\times$ 50 steps

Core parameter grids

Initial pollution: 10%, 30%, 50%
Coverage, timing and symmetric accuracy
Untreated, legal and platform conditions

Global sensitivity design

500 Latin hypercube vectors per regime
Reused across initial pollution levels
Legal and platform inputs sampled separately

Supplementary experiments

Targeting and constrained review capacity
Responsive workload; asymmetric TPR/TNR
Network topology and size

B Initialize each run

New seed and network per run

BA benchmark (static)

Undirected; $N = 1,000$; $m = 2$



Illustrative network

Superspreaders: top 1%
of accounts by degree

Initial pollution: 10%, 30%, 50%
Uniform sampling
without replacement

Initialization only: $t = 0$

C Within-run update

One fixed regime per run; steps $t = 1, \dots, 50$

1 Diffuse pollution

Step-start polluted accounts try to pollute clean neighbors.
Newly polluted accounts transmit from the next step.

No governance

No review

Legal intervention

$t = 1 + j\tau_L$

Platform moderation

Every step

not due

due

2 Set workload, select and classify

Fixed workload; state-responsive platform variant
Select from all accounts: degree-based or random
TPR/TNR: TP cleans; FP records wrongful intervention

3 Record post-update states and review outcomes

$t < 50$: next step, including after zero pollution

$t = 50$

D Run-level outcomes

Summarized at the end of each run

Mean effectiveness, $\bar{E}$

Mean clean-account percentage
over steps 1-50
Initialization excluded

Wrongful exposure, $O(50)$

Distinct accounts with at least
one false-positive intervention
by step 50 (% of all accounts)

Additional diagnostics

Precision; cumulative reviews;
TP / FP / TN / FN counts;
final effectiveness

E Across-run analysis and hypothesis assessment

Analyze each initial pollution level separately

H1 Parameter interactions

Grid-based finite-change contrasts
95% intervals for means and contrasts

H2 Relative parameter sensitivity

Primary: PAWN
Supplementary: partial rank correlations
Bootstrap, bin and replication checks

H3 Effectiveness-exposure trade-offs

Joint changes in effectiveness and exposure
Expansion and classification-quality contrasts